

README - Trinary Processor & OS Project

Trinary Processor & Operating System

Overview

The **Trinary Processor (T-CPU)** and **TrinaryOS (T-OS)** project emulates a balanced ternary computing architecture. It operates natively on vectors of (-1, 0, +1), representing **IRRELEVANT (-1)**, **UNKNOWN (0)**, and **RELEVANT (+1)** states. This system demonstrates how ternary logic can be applied to hardware emulation, operating system design, and process scheduling.

Features

- **TrinaryProcessor:** A 14-node balanced ternary processor supporting AND, OR, and refract-rotation operations.
- **TrinaryOS:** A kernel that triages processes into Active, Shortfall, and Void queues using early-exit logic.
- **Wavefront Execution:** Processes run on ternary wave signatures, propagating through hardware registers.
- **Status Register:** Tracks system health, signaling RELEVANT, UNKNOWN, or IRRELEVANT states.

Runtime Behavior

Processor Example

```
Executing Strict AND Filtering into Accumulator...  
Accumulator Vector : [ 1. 0. 1. 1. 0. -1. 0. 1. -1. 0. 0. 1. 1. 0.]  
Hardware Status Code: 1 (Status: RELEVANT)
```

OS Example

Initializing Core OS Capabilities...

KERNEL: Process 'TelemetryWaveDecoder' [PID 100] flagged as RELEVANT. Scheduled to Active Queue.

KERNEL: Process 'QuantumStreamListener' [PID 101] flagged as UNKNOWN. Parked in Shortfall Standby.

KERNEL: Process 'CorruptedBufferGhost' [PID 102] flagged as IRRELEVANT. Routed to Void Queue.

--- OS Scheduler Cycle Execution ---

Active Tasks: 1 | Shortfall Standby: 1 | Void Cleaned: 1

[Purge Engine] Garbage collecting 1 dead-end threads to preserve registers.

-> Ability One running on wave shape: [-0. 0. -0.]...

KERNEL: Process 'TelemetryWaveDecoder' [PID 100] flagged as UNKNOWN. Parked in Shortfall Standby.

--- End of Scheduler Cycle ---

Project Goals

- Demonstrate balanced ternary computing principles.
- Explore alternative processor architectures.
- Provide a sandbox for experimenting with ternary logic in operating systems.
- Inspire research into non-binary computing paradigms.

Roadmap

- ☐ Expand instruction set (e.g., XOR, NOT).
- ☐ Implement compiler or assembler for T-CPU.
- ☐ Extend OS with multi-process scheduling.
- ☐ Add visualization tools for wavefront propagation.

Contributing

Contributions are welcome! Please see CONTRIBUTING.md for guidelines.

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Sponsorship

Support this project through GitHub Sponsors. Contributions help fund further development, documentation, and research into ternary computing.

Trinary computing is more than a curiosity-it's a glimpse into alternative architectures that challenge binary dominance.